

ALEYNA ALTUNSU

AI Engineer · LLMOps & Computer Vision

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PROFESSIONAL SUMMARY

AI Engineer with hands-on production experience in **LLMOps, Computer Vision, NLP, and ML system design**. I have taken end-to-end ownership across a broad spectrum of production projects — from multi-agent LLM systems and Vision Transformer-based industrial defect detection to on-premise RAG architectures, LLM fine-tuning, reinforcement learning, and edge AI.

I combine technical depth with research discipline and community leadership: I co-founded BAU AI Society from scratch and grew it into one of the university's largest student organizations. I move fast in adapting cutting-edge methodologies such as DSPy, Table-RAG, Deformable Attention, and Q-Former to production; I am equally at home in academic R&D and industrial deployment.

TECHNICAL SKILLS

Languages & Tools	Python (primary), C++, SQL, Arduino · Git/GitHub, Docker, Docker Compose, WSL2, Virtual Environments
LLM / Agent / RAG	LangChain, LlamaIndex, DSPy, CrewAI, OpenAI API, LiteLLM, Ollama, HF Transformers · RAG (basic & hierarchical), multi-agent orchestration, intent routing, prompt slicing, chain-of-thought, SFT fine-tuning
Fine-tuning	Unsloth (4-bit QLoRA, 2× speed), PEFT, LoRA, GPTQ/AWQ quantization, DeepSeek-R1 integration · Llama 3.3, Mistral, Phi-4, Qwen 2.5, GPT-4o, Claude (API)
LLMOps / MLOps	FastAPI, WebSocket, REST API, Pydantic v2 · Docker/Compose, on-premise deployment · MLflow (tracking, registry, versioning), Hyperopt · Prometheus metrics, retry mechanisms, ThreadPoolExecutor, hallucination detection
Computer Vision	YOLOv8/v10, Faster R-CNN, SegFormer (semantic seg.) · CLIP (zero-shot), ViT, Swin Transformer, Deformable Attention, Q-Former · OpenCV, DICOM processing, synthetic data generation, CUDA/PyTorch memory optimization
Deep Learning / ML	PyTorch, TensorFlow/Keras · CNN, RNN, LSTM, GRU, encoder-decoder, ResNet, DenseNet, MobileNet · Transfer learning, ensemble learning, cross-validation · Random Forest, SVM, XGBoost, Logistic Regression
NLP	TF-IDF, Word2Vec, FastText, sentence-transformers · Text classification, sentiment analysis, document QA, PDF information extraction, multilingual evaluation
Data & Databases	Pandas, NumPy, scikit-learn pipelines, EDA, feature engineering · PostgreSQL (3NF, JOIN, full-text search), MongoDB, FAISS · Matplotlib, Seaborn
Edge AI / Other	Arduino, ESP32, PIR & DS18B20 sensor integration · Qiskit, Quantum ML · React/Vite (AI dashboard), WebSocket-based live UI

WORK EXPERIENCE

AI Engineer · Heysem AI, Eskişehir

2026

- ERP Financial AI Assistant: Built a multi-agent system that converts Turkish natural language queries into SAP B1 OData calls, generating balance sheet / aging / depreciation / segmentation reports, interactive charts, PDF output, and automated email delivery.

- Multi-Agent Architecture (CrewAI): Orchestrated reader / transaction / reporting layers with regex-based intent routing; all ERP entities, formulas, and NL→OData patterns managed via a single YAML configuration.
- Performance: Intent-aware prompt slicing reduced token count by ~70%. Response latency dropped from 2–4 min → 15–30 sec. Deterministic Python tool layer (@meta fields) eliminated LLM-induced numerical errors.
- Reliability: Server-side hallucination detection (regex), urllib3 retry layer, automatic 401/-8012 session recovery, parallel SAP queries via ThreadPoolExecutor.
- Industrial Defect Detection: 2-stage hybrid pipeline — Stage 1: CLIP semantic classification; Stage 2: SegFormer pixel-level segmentation. Custom defect datasets and synthetic data generation improved model robustness.
- Research: Studied Deformable Attention and Q-Former mechanisms in preparation for next-generation multimodal system architecture.

AI & Data Science Intern · Huawei R&D Center

2025

- Contributed to architectural analysis of Dify-like AI workflow platforms; observed production-ready LLM infrastructure design principles first-hand.
- Structured Prompt Programming with DSPy: Converted free-text prompts into DSPy modules to make pipelines interpretable and auto-optimizable; conducted advantage/limitation analysis versus traditional prompt engineering.
- Ran multilingual LLM evaluation tests across 3+ languages including Turkish; compiled findings into internal research reports.
- Investigated Table-RAG (2025) technique for structured-data RAG integration applicability. (Under NDA)

AI Engineer Intern — NLP · Enqura

2024

- Benchmarked TF-IDF, BoW, Word2Vec, and FastText representations on an Amazon review dataset; systematically documented strengths and weaknesses of each approach.
- Developed an LSTM-based 3-class sentiment analysis model; improved accuracy significantly via lemmatization & stop-word removal pipeline.
- Conducted LLM fine-tuning & pre-training experiments; built a PDF document QA system with PyMuPDF + FAISS + LangChain.
- Published 3 Medium articles; developed an interactive word prediction game with 3D embedding visualization.

AI Intern — Project Management · SolPro Ltd.

2024

- Contributed to the development of ML-based predictive maintenance systems for polymer production optimization and quality control.
- Built a market analysis automation chatbot using Gemini API: automated response generation, personalized customer email drafting, and trend visualization.

AI Intern — Computer Vision · Heysem AI

2024

- Optimized object detection and face recognition models using YOLOv8 & YOLOv10; halved training epochs while maintaining accuracy metrics.
- Built a real-time face recognition-based automated attendance system for security cameras; implemented voiceover integration on edge devices.

FEATURED PROJECTS

SAP B1 Financial AI Assistant *Python 3.11 · FastAPI · CrewAI · LiteLLM · WebSocket · React/Vite · Docker · Prometheus*

- Multi-agent AI assistant converting Turkish natural language to OData; delivers balance sheet, aging, depreciation, segmentation reports, PDF output, and email delivery.
- ~70% token reduction via intent-aware prompt slicing. Deterministic Python tool layer zeroed out hallucinations. Latency: 2–4 min → 15–30 sec.

Industrial Defect Detection — 2-Stage Vision Transformer Pipeline *PyTorch · CLIP · SegFormer · ViT · Swin Transformer · CUDA*

- Stage 1 CLIP zero-shot semantic classification + Stage 2 SegFormer pixel segmentation; synthetic data generation improved generalization on rare defect types.
- Optimized GPU memory consumption during high-resolution inference via CUDA/PyTorch compatibility layer.

KVKK-Compliant Legal Document Analysis — On-Premise RAG *FastAPI · React · FAISS · Ollama · PyMuPDF · sentence-transformers · Docker*

- Fully on-premise RAG pipeline with zero data leaving the organization. Section-aware hierarchical chunking minimized context loss compared to fixed-size chunking.
- Map-reduce summarization handled LLM context window overflow; Docker Compose enabled zero-external-dependency reproducible deployment.

SQL vs NoSQL Performance Benchmark *Python · PostgreSQL · MongoDB · Docker Compose · Matplotlib*

- Realistic e-commerce scenario: 10K users, 2K products, 30K orders. Automated benchmark across 8 operation types. Aggregation: PostgreSQL 26ms vs MongoDB 1,691ms (65× difference).

LLM Fine-tuning — Efficient Training with Unsloth *Python · PyTorch · Triton · Hugging Face · Unsloth*

- Fine-tuned Llama 3.3, Mistral, Phi-4, and Qwen 2.5 with QLoRA; 80% less GPU memory and 2× speed vs. standard training. DeepSeek-R1 chain-of-thought integration.

Pneumonia Detection — Deep Learning *TensorFlow · ResNet50V2 · DenseNet121 · Transfer Learning*

- Binary classification on 26,000+ DICOM chest X-rays. Patient-level stratified split prevented data leakage. ResNet50V2 + DenseNet121 ensemble achieved AUC 0.8755.

Graph-Based Behavioral Recommendation System — LLM vs Non-LLM *Python · Node2Vec · PyTorch · Graph Theory*

- Modeled view, click, and transition interactions as a user-item graph; generated embeddings with Node2Vec.
- Compared LLM-based semantic matching vs. behavioral graph approach via isolated experimental design; documented the advantages of the graph-based pipeline for production scalability.

AI-Powered Smart Heating System — Graduation Project *Python · ESP32 · scikit-learn · PIR & DS18B20 · IoT*

- Real-time human detection and heating automation via ESP32, PIR motion sensor, and DS18B20 temperature sensor integration.
- Delivered a working physical prototype combining AI decision-making with hardware control.

Real-Time Object Detection, Face Recognition & Automated Attendance *Python · YOLOv8/v10 · R-CNN · OpenCV · Edge Devices*

- Real-time object detection on security camera footage with YOLOv8 & YOLOv10; halved training epochs while preserving accuracy metrics.
- Face recognition-based attendance system: detected faces logged to database with timestamps, unknown entries flagged separately. Voiceover integration on edge device.

Earthquake Magnitude Prediction — MLflow Lifecycle *Python · MLflow · scikit-learn · Hyperopt · Flask*

- End-to-end ML lifecycle management with MLflow on Kandilli Observatory seismic data: experiment tracking, model registry, versioning, and Flask-based deployment.
- Optimized Random Forest Regressor with Hyperopt; systematically documented the hyperparameter search process.

Lane Detection — CLR/.NET Reimplementation *C++ · CLR/.NET · OpenCV · Computer Vision*

- Reimplemented a lane detection system for autonomous vehicle navigation in a CLR/.NET environment; optimized real-time performance and accuracy.

Multimodal AI: Image Classification & Multilingual Translation *Python · PyTorch · MobileNetV2 · Hugging Face M2M100*

- Combined MobileNetV2 image classification and M2M100 multilingual text translation in a single pipeline; resolved class mapping, tokenization, and model latency issues for production-ready integration.

6-Class Image Classification with CNN *Python · TensorFlow · CNN · Data Augmentation*

- Custom CNN architecture for 6 categories (man, woman, girl, boy, cat, dog); 85% validation accuracy. F1 score ranged 0.79–0.87; analyzed class imbalance and hyperparameter improvement paths.

Document QA System with FAISS & LangChain *Python · FAISS · LangChain · sentence-transformers · PyMuPDF*

- Extracted text from PDFs via PyMuPDF & regex, generated embeddings with Sentence-Transformers, built efficient dense retrieval with FAISS indexing. Context-aware QA system via LangChain retriever integration.

Amazon & IMDB Sentiment Analysis *Python · scikit-learn · TF-IDF · Word2Vec · LSTM · NLP*

- Benchmarked TF-IDF, Word2Vec, and BoW for Amazon review classification (positive/negative/neutral); optimized preprocessing pipeline with lemmatization & stop-word removal.
- NLP-based movie review classification on IMDB dataset; evaluated contribution of results to recommendation systems.

Word Prediction Game with 3D Embedding Visualization *Python · Word2Vec · TF-IDF · FastText · BoW · Neural Network*

- Inspired by Contexto.me: interactive word prediction game that visualizes word embeddings in 3D space. Conducted comparative performance analysis of Word2Vec, TF-IDF, FastText, and BoW; improved prediction accuracy with a neural network model.

Cross-Validation & Hyperparameter Optimization — MNIST *Python · scikit-learn · SVM · GridSearchCV · RandomizedSearchCV*

- Optimized SVM classifier with GridSearchCV and RandomizedSearchCV; improved accuracy from 94.64% → 96%. Prevented overfitting with 5-fold cross-validation.

Stroke Risk Prediction *Python · scikit-learn · Random Forest · EDA*

- Identified relevant features through EDA on a health dataset; developed a stroke risk prediction model using Random Forest and other classification algorithms.

AI-Powered Market Analysis Chatbot *Python · Gemini API · Generative AI · Data Visualization*

- Built a market analysis automation chatbot for SolPro using Gemini API: automated response generation, personalized customer email drafting, and trend visualization. Hands-on experience converting a general-purpose LLM into a domain-specific business tool.

Self-Solving Pacman — Reinforcement Learning *Python · A* · Q-Learning · Reinforcement Learning*

- AI agent that plays Pacman: implemented A* search for shortest-path maze navigation and Q-Learning for reward-based learning. Compared the effect of different reward functions on agent strategy.

EDUCATION

B.Sc. in Artificial Intelligence Engineering · [Bahçeşehir University \(BAU\)](#) 2021 – 2025

- Co-founder of BAU AI Society (BAU AIS); grew it into one of the university's most active and largest clubs during my presidency.
- Academic papers on AI ethics, singularity, leadership, and societal applications of artificial intelligence.

BILSEM — Science and Arts Center (MoNE) · [Eskişehir Emine Emir Şahbaz](#) 2013 – 2021

- Passed the gifted individuals exam in 5th grade; received 8 years of intensive education in science, robotics, philosophy, mathematics, and fine arts.
- Member of RoboEs robotics team — participated in national and international competitions.

VOLUNTEERING & COMMUNITY

President · **BAU AIS** 2023 – 2024

- Elevated the society to one of the university's largest student clubs. Established sponsorship, organization, and board sub-teams, transforming it into a structured institution. Organized multiple large-scale events including International Future AI Summit '24.

Co-Founder & Vice President · **BAU AIS** 2022 – 2023

- Founded the society from scratch; managed member acquisition, event coordination, internal communication infrastructure, and external partnerships.

Board of Auditors · BAU AIS

2024 – 2025

- Served as advisor and auditor for 10 months to ensure continuity of club culture and institutional memory.

Social Media Head · TRYZKB (SSB Affiliate)

2023

- Developed social media strategy for a national AI club network affiliated with the Turkish Defence Industries Presidency (SSB); built the organization's corporate digital presence from scratch.

Committee Member · IEEE Educational Activities & IEEE Computer Society

2021 – 2023

- Organized events introducing university students to computer science career paths; built network with startup founders.

CERTIFICATIONS

2025 MLOps Platforms: Amazon SageMaker & Azure ML (Coursera – Duke University) · Structuring ML Projects (DeepLearning.AI) · Image Processing (x2) (Coursera – MathWorks) · Image & Video Processing (Coursera – Duke University)

2024 Improving Deep Neural Networks · Quantization Fundamentals · Finetuning LLMs · Pretraining LLMs (DeepLearning.AI) · AI Principles w/ Edge Computing · IBM ML Specialization · Intro to Embedded ML · IoT System Design (Coursera)

2023 Neural Networks & Deep Learning (DeepLearning.AI) · Data Cleaning / Feature Engineering / Geospatial Analysis / Intro to ML / Intro to SQL (Kaggle)

LANGUAGES

Turkish (Native) · **English** (Advanced — technical & academic) · **French** (Beginner)

References available upon request.